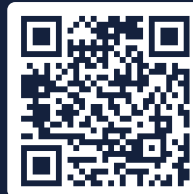


Bosuk Nam 남보석

Backend Developer · Java / Spring

Infra Development Team · Kakao Corp (Mar 2022 - Present)



Portfolio site
bosuknam.github.io

SUMMARY

Backend developer who builds internal systems for large-scale infrastructure end-to-end, from design to operations. At Kakao I have spent 4 years 7 months developing and operating the server management, idle-server reclamation and company-wide governance systems that cover roughly **230K servers (PM + VM)** and **4,000+ users**.

I led the design and implementation of the idle-server management system single-handedly.

At LG CNS I was a framework developer on the in-house Spring framework (DevOn), owning dev environments, guides and training for MSA projects at major banks.

I turn repetitive operations into batches, alerts and CI/CD.

Currently I focus on embedding n8n and agentic AI into the daily development loop (ticket → implement → review → PR → docs).

9y 7m

Experience

230K+

Servers managed

4,000+

System users

10+

Projects

HIGHLIGHTS

- Systemized the full idle-server flow — **detect** → **notify** → **approve** → **return** — and designed an org-lead approval process to raise the return rate
- Introduced **SLOs for batch jobs**: automatic alert when runtime exceeds mean + 3σ , catching batch anomalies before people do
- Turned recurring CS requests (e.g. bulk mapping changes) into **web/API features instead of admin DB queries** — Excel upload with real-time status
- Automated resignee detection and pending-approval alerts, eliminating manual lookups by administrators
- Framework developer for LG CNS's DevOn Boot; owned dev standards, guides and training on KB Kookmin Bank and Shinhan Bank MSA projects

CORE STACK

Java

Spring Boot

Spring Batch

JPA / MyBatis

MySQL / MongoDB

Redis

Kubernetes

Docker

Jenkins / ArgoCD

PromQL

Node-RED

n8n

Agentic AI

CAREER TIMELINE

Mar 2022 - Present	Kakao Corp	Backend Developer — server, idle-server & governance systems
Jul 2017 - Mar 2022	LG CNS	Senior Engineer — IIoT, pharma MES, DevOn Framework, banking MSA
Apr 2017 - Jun 2017	TmaxSoft	Technical support (ProObject, AnyLink)
Jan 2015 - Apr 2016	Kunhwa Engineering	Urban planner

Backend development and operations for the internal systems that manage Kakao's infrastructure assets — about 230K physical, virtual, K8S and provisioning servers.

Used by ~4,000 people at HQ plus affiliates such as Kakao Enterprise, Kakao Mobility and AXZ.

Infra Idle Server Management System

Sole lead · design / build / operate

Mar 2022 - Present

CONTEXT Among 230K servers, unused and under-used machines were leaking cost.

Finding owners and reclaiming them was manual work.

- WORK**
- Batch that syncs company-wide server metrics (PromQL) to **automate idle detection**; weekly personal alerts to owners plus department-level statistics
 - Designed and built an **org-lead approval process** to address the low return rate (rolled out 2026, impact under review)
 - Open API key request/issuance for external integration and user management (JWT, Redis)
 - Cron scheduling and a monitoring dashboard for hourly (2h/5h), daily and weekly batches; **failure alerts + SLO (3σ) alerts** via Node-RED
 - CI/CD automation on K8S · Docker · SonarQube

OUTCOME **Every step of idle-server reclamation is now system-driven. Batch anomalies are detected automatically by the 3σ rule, minimizing operator intervention.**

STACK Spring Boot · REST API · JWT · Redis · Spring Batch · Node-RED · RDBMS/SQL · PromQL · K8S · Docker · SonarQube

Company-wide Governance System

Design lead · development

Feb 2025 - Present

CONTEXT Project, cost and owner data were scattered, so administrators queried the DB by hand.

Exceptions such as resignations or pending approvals surfaced late, and recurring CS requests were handled with ad-hoc queries.

- WORK**
- Company-wide project and **cost dashboard** plus owner management; change history tracked with Envers
 - Batches for per-project server mapping sync, **resignee detection and pending-approval alerts** (RDBMS + MongoDB, JPA)
 - Converted recurring CS work into **web/API features** — bulk mapping changes via Excel upload, with change status visible in real time
 - User management and statistics via Open API key issuance (REST API, JWT)
 - CI/CD on K8S · Docker · SonarQube

OUTCOME **Replaced administrators' manual DB queries with web features, batches and alerts — the system now surfaces exceptions first.**

STACK Spring Boot · REST API · JWT · Spring Batch · RDBMS · MongoDB · Envers · JPA · K8S · Docker · SonarQube

Infra Server Management System

Development · operations

Mar 2022 - Apr 2025

CONTEXT Status and owner data for servers deployed at HQ and affiliates frequently drifted from reality.

Incident response had no documented procedure.

- WORK**
- Server management system for HQ and affiliates (Web / REST API, MSA)
 - **Server integrity check & alerting** and owner-refresh batches (Spring Batch, Redis)
 - Wrote the incident response process guide (Wiki)
 - CI/CD with Jenkins · K8S · Docker · SonarQube

OUTCOME **Server data integrity is continuously verified by batch; incident response is standardized.**

STACK Spring Boot · REST API · MSA · Spring Batch · Jenkins · RDBMS/SQL · Redis · K8S · Docker · SonarQube

I treat automating repetitive work as part of the job. These workflows are in daily use.

Agentic AI development pipeline

Jira ticket creation →
implementation → AI code review →
PR authoring → incorporating review
feedback, run as an agent workflow.
Design docs in Discussions/Wiki and
HTML presentation decks are
generated routinely as well.

n8n operations automation

Internal Q&A bot, failure handling
with automatic retries for batches
and jobs, auto-filling values on
internal sites.

Plugin / Skills development

Agent plugins and skills tailored to
the team's workflow to standardize
recurring procedures.

LG CNS

DevOn Solution Team · Senior Engineer

Jul 2017 - Mar 2022 (4y 9m)

Started as a smart-factory solution developer, then moved to the team building the in-house Spring framework (DevOn Framework / DevOn Boot).

Served as framework developer on MSA projects for major banks — owning dev environments, standards and training.

Won the new-hire project grand prize; taught Java as an internal instructor.

Shinhan Bank SOL New Services (MSA)

Nov 2021 - Mar 2022

Framework developer

Design phase of a project building non-banking features (baseball, coupons, events) inside the SOL app as microservices. Owned dev environment setup, guides, training and technical support.

- WebClient-based sync / async / parallel call module (@Async, CompletableFuture)
- Redis · RDBMS · Kafka integration, Swagger, per-profile configuration
- HyperCloud K8S environment

Java 8 · Spring Boot · Kafka · Redis · Gradle · GitLab

KB Kookmin Bank Teen E-Wallet (MSA)

Apr 2021 - Nov 2021

Framework developer

Prepaid e-wallet app for users under 18 (deposit/withdraw, transfers, allowance requests). Microservices on AWS with K8S/Docker.

- JWT (header ↔ Redis) auth, LogAspect logging customization, multiple datasources
- **Netty-based TCP module** for EAI integration
- Emoji storage incident resolved with utf8mb4 + HikariCP configuration

Java 8 · Spring Boot/Batch · AWS · K8S · Kafka · Redis · Netty · Maven

DevOn Framework / DevOn Boot

Sep 2020 - Mar 2022

Framework developer

Feature development, bug fixes and releases for the in-house Spring-based framework, plus project technical support, training and proposals.

- Added WebClient · JWT · Swagger2 · Netty(TCP) modules; migrated Maven → Gradle
- WebFlux REST · Thymeleaf · Spring Batch templates; Jenkins → Nexus releases
- Log4Shell (Log4j) response; proposals for Shinhan and KB Kookmin Bank

Spring Framework/Boot · WebFlux · Jenkins · Nexus · Jira · Confluence

MES Core Platform (LG group-wide)

Apr 2020 - Jun 2020

[REST API server](#)

Group-wide smart-factory SPA platform supporting legacy compatibility and new extensions.

- JWT session / user management; JPA with MongoDB and MSSQL
- Solace pub-sub messaging; legacy integration via BizActor interface

Spring Boot · JWT · JPA · Solace · BizActor

iPharmMES HTML5 Migration

Jul 2019 - Sep 2020

[Backend / Frontend](#)

Migrated a Silverlight-based pharmaceutical MES to Spring Boot + JSP (MVC Model 2).

- Spring Security roles/sessions, SSO, i18n, TLD permission tags
- Silverlight ↔ JSP hybrid (iframe), IE/Chrome compatibility

Spring Boot · Spring Security · MSSQL · BizActor · JSP

Industrial IoT Platform

Sep 2017 - Dec 2018

[Admin FE/BE](#) · [on-site deployment](#)

IIoT platform collecting and managing high-volume factory equipment data (Storm · Spark · HBase · Kafka · Redis · Node-red).

- Real-time Kafka data pushed to screens over WebSocket; admin UI (device registration, control, logs, screen designer)
- Deployed and operated at LG Display Paju, LG Chem Ochang and CJ Logistics sites
- Grid rendering performance and eZ-UMS memory (RUI → AUI) improvements

Spring MVC · Kafka · Redis · HBase · Node-red · Mule ESB · quartz

Internal Java Instructor for New Hires

May 2019 - Aug 2019

[Instructor](#)

Java fundamentals: JVM, collections, exceptions, threads, file IO.

- Satisfaction **6.88 / 7.0** (SW Boot Camp), **6.90 / 7.0** (IT Career Start Technology)

Java 8

SIDE PROJECTS

lottery	Rust	Generates lottery number combinations from historical draw data. Web version at bosuknam.github.io/lottery github.com/bosuknam/lottery
kis-trading	Python	Tracks domestic/overseas stock trends and indicators and helps buy undervalued stocks (Korea Investment & Securities API) github.com/bosuknam/kis-trading
childcare-talk	Dart / Flutter	Mobile app for logging childcare conversations github.com/bosuknam/childcare-talk
background-sound-app	JavaScript	One-click background sounds for different situations github.com/bosuknam/background-sound-app

SKILLS

Language	Java · SQL · JavaScript · Python
Backend	Spring Boot · Spring Framework/MVC · Spring Batch · Spring Security · WebFlux/WebClient · JPA/Hibernate (Envers) · MyBatis · Netty · REST API · JWT · MSA
Data	MySQL · MariaDB · Oracle · MSSQL · AuroraDB · MongoDB · Redis · HBase
Messaging	Kafka · Solace
Infra / DevOps	Kubernetes · Docker · Jenkins · GitHub Actions · ArgoCD · SonarQube · Nexus · AWS · Prometheus/PromQL · Node-RED · Airflow
AI / Automation	Agentic AI workflows · Plugin/Skills development · n8n
Collaboration	Git · GitLab/Bitbucket · Jira · Confluence · Wiki · Swagger
Frontend (secondary)	JSP · Thymeleaf · HTML5/CSS · jQuery

EDUCATION · CERTIFICATIONS

2009 - 2015	B.S. Urban Planning & Engineering, Hanyang University
May 2014	Engineer Urban Planning — HRD Korea
2016 - 2017	Applied Software Convergence Project Course, Lectorpia (C/C++ · Java · Android · Spring)
Mar 2017	TOEIC Speaking 170 (Lv.7)
Aug 2018	Neural Networks and Deep Learning — Coursera
Nov 2019	Engineer Information Processing — HRD Korea

EARLIER CAREER · ETC.

May 2011 - Feb 2013	Military — Republic of Korea Army, sergeant, honorably discharged
Jan 2015 - Apr 2016	Kunhwa Engineering — urban planner (Paju Unjeong 3 district development, Juan redevelopment)
Apr 2017 - Jun 2017	TmaxSoft — technical support (ProObject, AnyLink). Left after receiving the LG CNS offer



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<https://bosuknam.github.io/>